Nmap Basic Port Scans

TCP and UPD Ports

Can classify ports in two states:

1. Open port indicates that there is some service listening on that port.
2. Closed port indicates that there is no service listening on that port.

However, in practical situations, we need to consider the impact of firewalls. For instance, a port might be open, but a firewall might be blocking the packets. Therefore, Nmap considers the following six states:

1. **Open**: indicates that a service is listening on the specified port.
2. **Closed**: indicates that no service is listening on the specified port, although the port is accessible. By accessible, we mean that it is reachable and is not blocked by a firewall or other security appliances/programs.
3. **Filtered**: means that Nmap cannot determine if the port is open or closed because the port is not accessible. This state is usually due to a firewall preventing Nmap from reaching that port. Nmap's packets may be blocked from reaching the port; alternatively, the responses are blocked from reaching Nmap's host.
4. **Unfiltered**: means that Nmap cannot determine if the port is open or closed, although the port is accessible. This state is encountered when using an ACK scan `-sA`.
5. **Open|Filtered**: This means that Nmap cannot determine whether the port is open or filtered.
6. **Closed|Filtered**: This means that Nmap cannot decide whether a port is closed or filtered.

TCP Flags

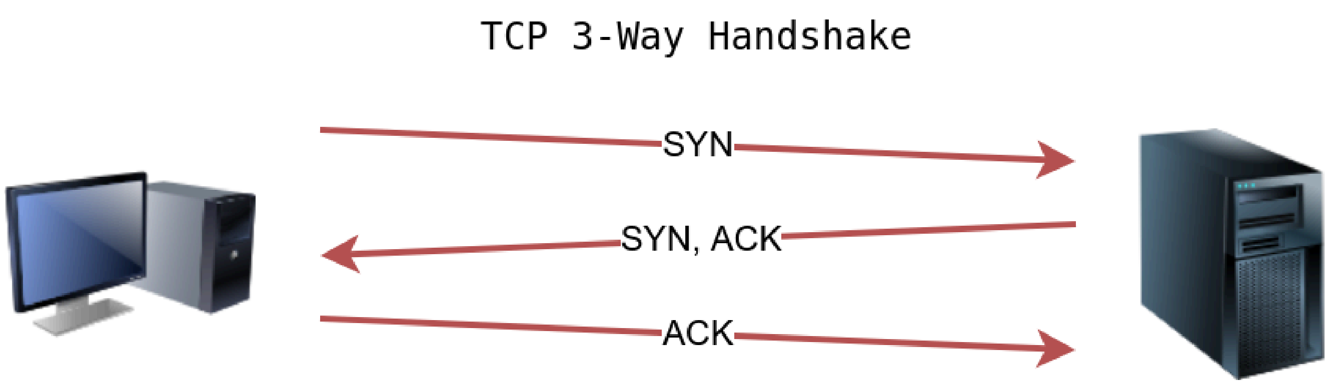
TCP header flags are:

1. **URG**: Urgent flag indicates that the urgent pointer filed is significant. The urgent pointer indicates that the incoming data is urgent, and that a TCP segment with the URG flag set is processed immediately without consideration of having to wait on previously sent TCP segments.
2. **ACK**: Acknowledgement flag indicates that the acknowledgement number is significant. It is used to acknowledge the receipt of a TCP segment.
3. **PSH**: Push flag asking TCP to pass the data to the application promptly.
4. **RST**: Reset flag is used to reset the connection. Another device, such as a firewall, might send it to tear a TCP connection. This flag is also used when data is sent to a host and

there is no service on the receiving end to answer.
5. **SYN**: Synchronize flag is used to initiate a TCP 3-way handshake and synchronize sequence numbers with the other host. The sequence number should be set randomly during TCP connection establishment.
6. **FIN**: The sender has no more data to send.

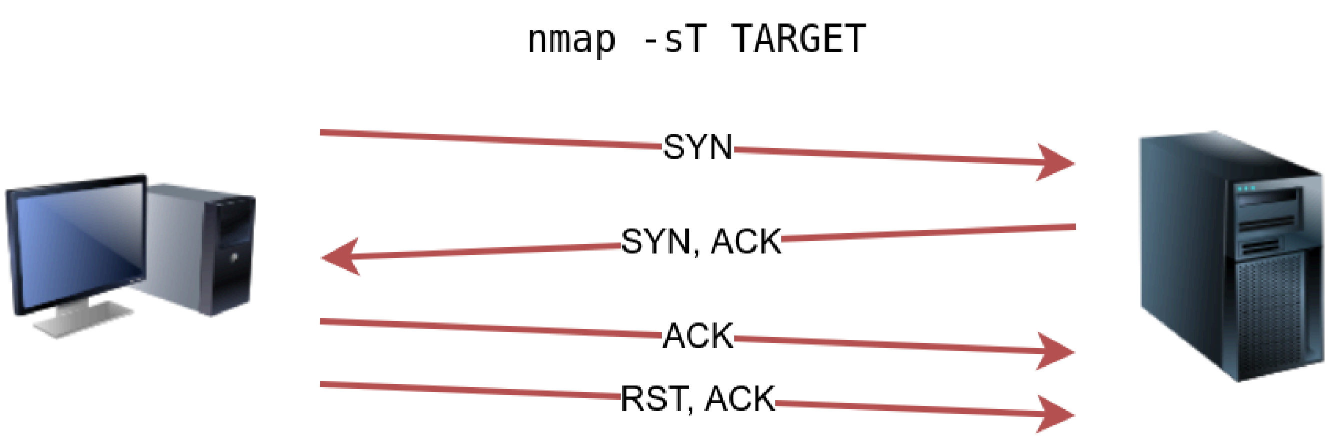
TCP Connect Scan

the client sends a TCP packet with SYN flag set, and the server responds with SYN/ACK if the port is open; finally, the client completes the 3-way handshake by sending an ACK.



Case: TCP port is open.

We are interested in learning whether the TCP port is open, not establishing a TCP connection. Hence the connection is torn as soon as its state is confirmed by sending a RST/ACK. You can choose to run TCP connect scan using `-sT`.



Case: TCP port is open.

It is important to note that if you are not a privileged user (root or sudoer), a TCP connect scan is the only possible option to discover open TCP ports.

By default, Nmap will attempt to connect to the 1000 most common ports

A closed TCP port responds to a SYN packet with RST/ACK to indicate that it is not open. This pattern will repeat for all the closed ports as we attempt to initiate a TCP 3-way handshake with them.

open ports would reply with SYN/ACK in wireshark

```
root@ip-10-128-67-78:~# nmap -sT 10.128.175.39
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-20 19:07 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or
ecify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disab
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.175.39
Host is up (0.0017s latency).
Not shown: 992 closed ports
PORT    STATE SERVICE
22/tcp  open  ssh
25/tcp  open  smtp
80/tcp  open  http
110/tcp open  pop3
111/tcp open  rpcbind
143/tcp open  imap
993/tcp open  imaps
995/tcp open  pop3s
```

`-sT` flag returns a detailed list of the open ports

`-F` to enable fast mode and decrease the number of scanned ports from 1000 to 100 most common ports.

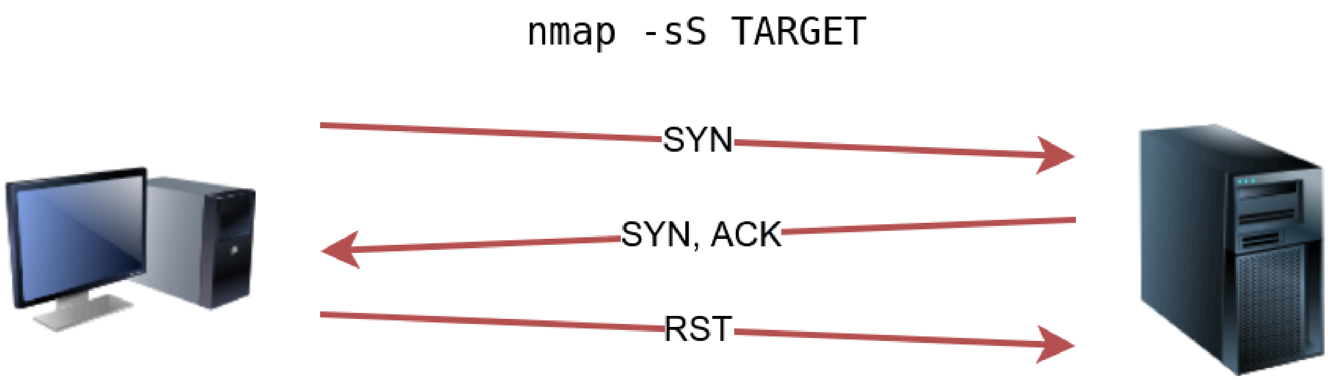
the `-r` option can also be added to scan the ports in consecutive order instead of random order. This option is useful when testing whether ports open in a consistent manner, for instance, when a target boots up.

TCP SYN Scan

SYN scan, and it requires a privileged (root or sudoer) user to run it

SYN scan does not need to complete the TCP 3-way handshake; instead, it tears down the connection once it receives a response from the server

select this scan type by using the `-sS` option



Case: TCP port is open.

TCP SYN scan is the default scan mode when running Nmap as a privileged user, running as root or using sudo, and it is a very reliable choice

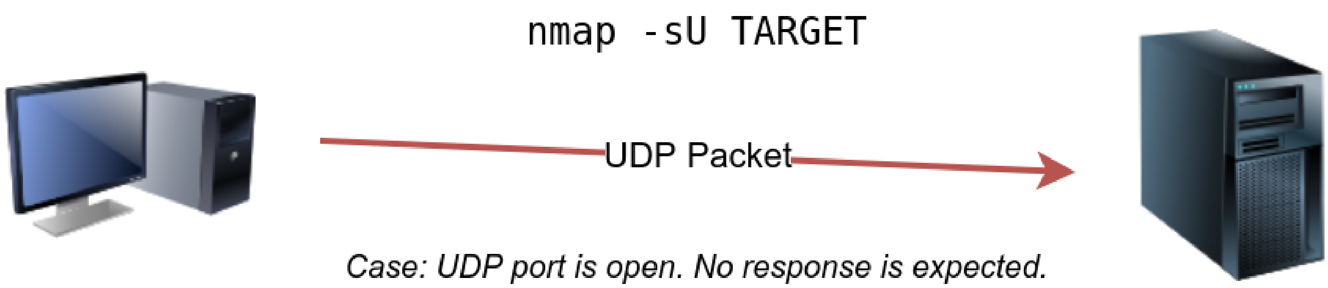
```
root@ip-10-128-67-78:~# sudo nmap -sS 10.128.173.88
sudo: unable to resolve host ip-10-128-67-78: Name or service not known
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-20 19:25 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or sp
ecify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.173.88
Host is up (0.0018s latency).
Not shown: 991 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
25/tcp   open  smtp
80/tcp   open  http
110/tcp  open  pop3
111/tcp  open  rpcbind
143/tcp  open  imap
993/tcp  open  imaps
995/tcp  open  pop3s
6667/tcp open  irc

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
```

UDP Scan

UDP is a connectionless protocol, and hence it does not require any handshake for connection establishment

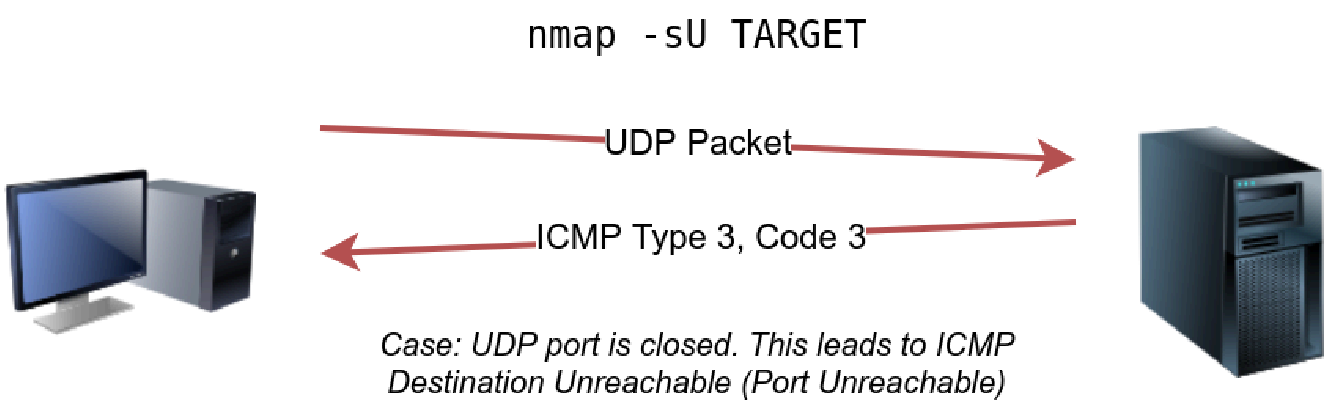
if a UDP packet is sent to a closed port, an ICMP port unreachable error (type 3, code 3) is returned

Can select UDP scan using the `-sU` option; moreover, you can combine it with another TCP scan.



sending a UDP packet to an open port won't tell us anything. we cannot expect any reply in return

UDP ports that don't generate any response are the ones that Nmap will state as open.



nmap -sU -F -v 10.128.138.198

```
root@ip-10-128-67-78:~# nmap -sU -F -v 10.128.138.198
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-20 19:34 BST
Initiating Ping Scan at 19:34
Scanning 10.128.138.198 [4 ports]
Completed Ping Scan at 19:34, 0.04s elapsed (1 total hosts)
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-ser
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
id servers with --dns-servers
Initiating UDP Scan at 19:34
Scanning 10.128.138.198 [100 ports]
Increasing send delay for 10.128.138.198 from 0 to 50 due to max_successful_tryno increase to 4
Increasing send delay for 10.128.138.198 from 50 to 100 due to max_successful_tryno increase to 5
Increasing send delay for 10.128.138.198 from 100 to 200 due to max_successful_tryno increase to 6
Increasing send delay for 10.128.138.198 from 200 to 400 due to max_successful_tryno increase to 7
Increasing send delay for 10.128.138.198 from 400 to 800 due to max_successful_tryno increase to 8
UDP Scan Timing: About 44.70% done; ETC: 19:35 (0:00:38 remaining)
Discovered open port 53/udp on 10.128.138.198
Discovered open port 111/udp on 10.128.138.198
Completed UDP Scan at 19:36, 101.03s elapsed (100 total ports)
Nmap scan report for 10.128.138.198
Host is up (0.00071s latency).
Not shown: 97 closed ports
PORT    STATE         SERVICE
53/udp  open          domain
68/udp  open|filtered dhcpc
111/udp open          rpcbind

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 101.17 seconds
           Raw packets sent: 223 (8.286KB) | Rcvd: 105 (6.602KB)
```

Fine-Tuning Scope and Performance

You can specify the ports you want to scan instead of the default 1000 ports. Specifying the ports is intuitive by now. Let’s see some examples:

- port list: `-p22,80,443` will scan ports 22, 80 and 443.
- port range: `-p1-1023` will scan all ports between 1 and 1023 inclusive, while `-p20-25` will scan ports between 20 and 25 inclusive.

You can request the scan of all ports by using `-p-`, which will scan all 65535 ports. If you want to scan the most common 100 ports, add `-F`. Using `--top-ports 10` will check the ten most common ports.

You can control the scan timing using `-T<0-5>`. `-T0` is the slowest (paranoid), while `-T5` is the fastest. According to Nmap manual page, there are six templates:

- paranoid (0)
- sneaky (1)
- polite (2)
- normal (3)
- aggressive (4)
- insane (5)

Note that `-T4` is often used during CTFs and when learning to scan on practice targets, whereas `-T1` is often used during real engagements where stealth is more important.

control the packet rate using `--min-rate <number>` and `--max-rate <number>`. For example, `--max-rate 10` or `--max-rate=10`

Summary

Port Scan Type	Example Command
TCP Connect Scan	`nmap -sT 10.128.138.198`
TCP SYN Scan	`sudo nmap -sS 10.128.138.198`
UDP Scan	`sudo nmap -sU 10.128.138.198`

Option	Purpose
`-p-`	all ports
`-p1-1023`	scan ports 1 to 1023
`-F`	100 most common ports
`-r`	scan ports in consecutive order
`-T<0-5>`	-T0 being the slowest and T5 the fastest
`--max-rate 50`	rate <= 50 packets/sec
`--min-rate 15`	rate >= 15 packets/sec
`--min-parallelism 100`	at least 100 probes in parallel